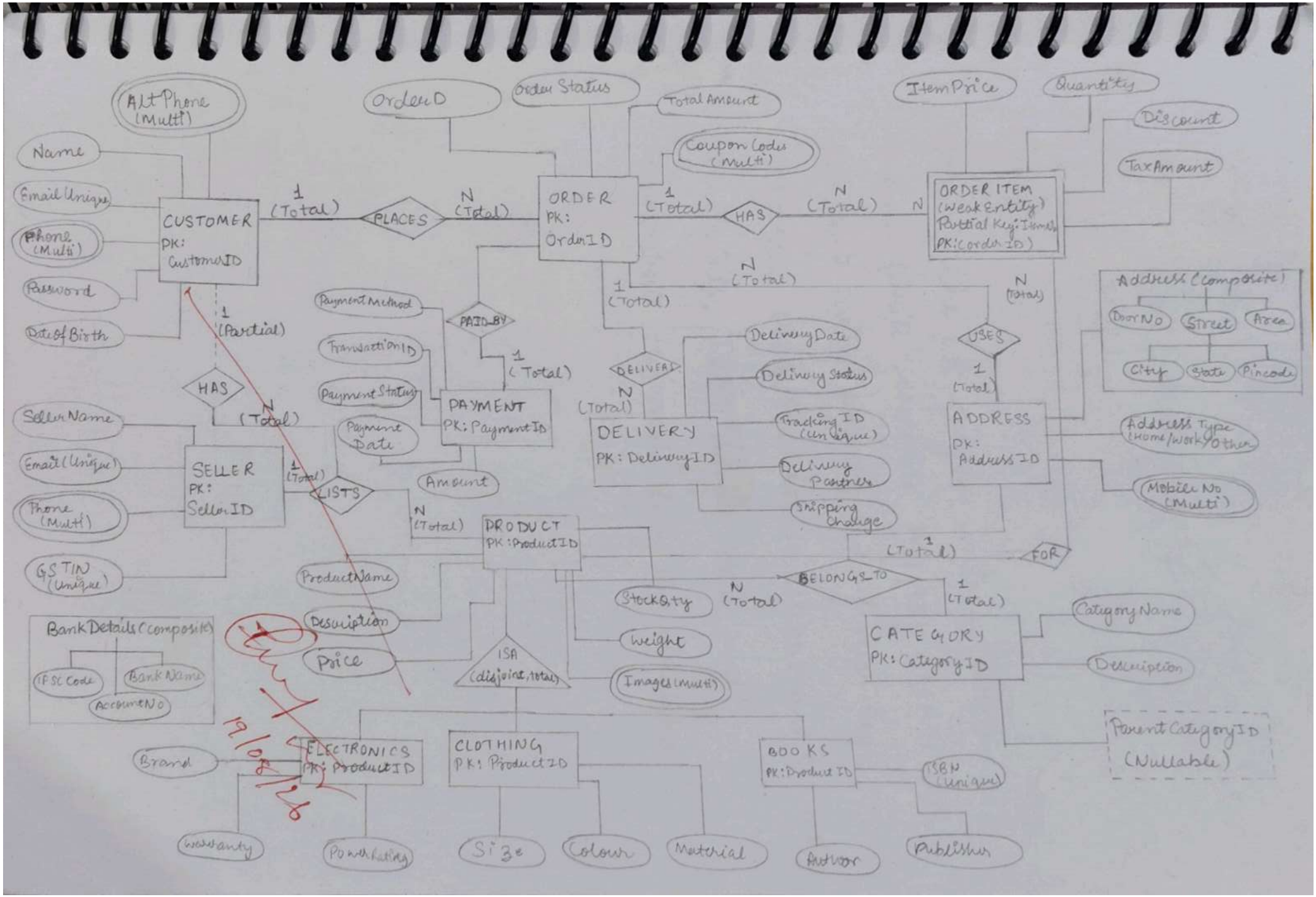
# Experiment - 1

## Aim:

Design an ER diagram for an Indian e-commerce platform with entities Customer, Product, Order, OrderItem, Seller, Category, Payment, Delivery, Address. Specify primary keys, composite and multi-valued attributes, weak entities, specialization of Product, and participation constraints.


CUSTOMER
PK: CustomerID
Name
Email Unique
Phone (Multi)
Password
Date of Birth
Alt Phone (Multi)
1 (Total)
PLACES
N (Total)
ORDER
PK: OrderID
OrderID
Order Status
Total Amount
Coupon Codes (Multi)
1 (Total)
HAS
N (Total)
ORDER ITEM
(Weak Entity)
Partial Key: Item No
PK: (OrderID)
Item Price
Quantity
Discount
Tax Amount
1 (Partial)
HAS
N (Total)
SELLER
PK: SellerID
Seller Name
Email (Unique)
Phone (Multi)
GSTIN (Unique)
1 (Total)
LISTS
N (Total)
PRODUCT
PK: ProductID
ProductName
Description
Price
Stock Qty
Weight
Images (Multi)
N (Total)
BELONGS_TO
1 (Total)
CATEGORY
PK: CategoryID
Category Name
Description
Parent CategoryID (Nullable)
FOR
1 (Total)
PAYMENT
PK: PaymentID
Payment Method
Transaction ID
Payment Status
Payment Date
Amount
PAID_BY
1 (Total)
DELIVERY
PK: DeliveryID
Delivery Date
Delivery Status
Tracking ID (Unique)
Delivery Partner
Shipping Charge
DELIVERS
N (Total)
1 (Total)
ADDRESS
PK: AddressID
Address (Composite)
Door No
Street
Area
City
State
Pincode
Address Type (Home/Work/Other)
Mobile No (Multi)
USES
1 (Total)
N (Total)
Bank Details (Composite)
IFSC Code
Bank Name
Account No
ISA (disjoint, total)
ELECTRONICS
PK: ProductID
Brand
Warranty
Power Rating
CLOTHING
PK: ProductID
Size
Colour
Material
BOOKS
PK: ProductID
ISBN (Unique)
Author
Publisher